

The Pest Risk Reduction Scenario Tool (PRReSTo) for quantifying trade-related plant pest risks and benefits of risk-reducing measures

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ABSTRACT

Plant pest risks posed by trade in agricultural commodities are managed through pest risk analysis which involves assessing risk and selecting phytosanitary measures to manage unrestricted risk if it is unacceptably high. Diverse measures can be used to reduce risk. However, calibrating risk management protocols with the level of unrestricted risk is challenging. Agreed methods and user-friendly tools for comparatively evaluating risk-reducing measures in different pest-host systems are lacking. Here, we describe the Pest Risk Reduction Scenario Tool (PRReSTo), a generic Bayesian network model that estimates likely infestation rates of host items in consignments. PRReSTo considers host items' vulnerability to pest infestation, the pest abundance to which host items are exposed during production, and the individual and cumulative effects of risk-reducing measures aimed at either minimising host vulnerability, pest exposure or infestation rates. We apply a PRReSTo implementation developed for assessing trappable insect pests in horticultural produce to test its versatility. First, we evaluate likely (unrestricted) host infestation rates under different levels of pest exposure and host vulnerability. Next, we evaluate the adequacy of risk-reducing measures under different host infestation scenarios and acceptability thresholds. We find that equivalent risk reduction may be achieved by highly effective individual measures or combinations of measures in a phytosanitary systems approach. The current implementation of PRReSTo is supported by an open access web application to facilitate adoption as a scenario analysis and decision-support tool by industry and regulators.

1. Introduction

Global trade in agricultural commodities poses significant plant pest risks. World Trade Organisation (WTO) members are required to manage those risks according to principles outlined in the *Agreement on the Application of Sanitary and Phytosanitary Measures* (WTO 1994). As a result, internationally agreed approaches to managing trade-related plant pest risks have been developed through the International Plant Protection Convention, European Food Safety Authority and other mechanisms (Devorshak 2012; Eschen et al., 2015; Jeger et al., 2018). The pest risk analysis (PRA) process involves pest risk assessment and pest risk management. Where unacceptable risks are identified through pest risk assessment, importing countries may require those risks to be managed through the application of diverse risk-reducing phytosanitary measures (IPPC 2007; Devorshak 2012; IPPC 2017c). Typically, risk-reducing measures have been classified according to when they are applied in the production and supply chain (Holt et al., 2018; Quinlan

et al., 2020). An alternative classification, or risk framework, was recently developed according to how measures reduce risk (van Klinken et al., 2020). It distinguished between four functional risk reduction objectives: minimise exposure to the pest, minimise vulnerability of the host, reduce pest infestation rates in the host, and reduce pest establishment (van Klinken et al., 2020). These objectives are consistent with the factors being considered in pest risk assessment as they pertain to the probability of pest entry and establishment, which is the focus of pest risk management (IPPC 2017c). Van Klinken et al. (2022) have extended their risk framework to apply to any traded commodity causing pest movement across borders (including both hosts and carriers) and developed a comprehensive "menu of [39 unique risk-reducing] measures" classified under the four objectives (van Klinken et al., 2022). A measure can be an officially mandated and monitored 'phytosanitary measure', or a 'commercial measure' that is a standard practice in the supply chain (e.g., an industry- or demand-driven quality specification that also limits the vulnerability of a host to pest infestation). Such

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commercially motivated measures may be considered when undertaking pest risk assessment (Griffin 2012).

Despite the diverse options available for managing plant pest risks, there remains little guidance on how to calibrate the design of risk management protocols with the level of unrestricted risk determined through pest risk assessment. For example, single-point treatments with demonstrated mortality rates of 99.99% (probit 8.7) or 99.9968% (probit 9.0) are commonly required, at least for high priority pests (IAEA 2011). However, such risk management requirements may not always consider biological or production factors that could already be limiting host infestation rates. For example, the infestation rate of fruit exposed to very high numbers of Tephritid fruit flies under laboratory or semi-natural conditions can vary significantly depending on host vulnerability (Follett et al., 2021). Similarly, pest abundance to which hosts are exposed during production can vary from zero to very high levels. Both host vulnerability and pest exposure can be reduced through the application of diverse measures, from growing less vulnerable varieties to managing, avoiding or excluding pests at the production site (van Klinken et al., 2022). Integrating both pest risk assessment and management is required for designing trade protocols that effectively reduce risks in a way that is least trade restrictive (Devorshak 2012; Favrin and Cree 2016). Quantitative models that incorporate the effect of risk-reducing measures into risk estimations while also accounting for uncertainty can provide a more informed basis for decision-making.

A wide range of quantitative approaches to PRAs have been developed, but these have often focused on specific pest-host systems or specific measures or sets of measures (Yamamura and Katsumata 1999; Hattingh et al., 2020). Agreed methods and user-friendly tools for evaluating diverse risk-reducing measures in the context of different pest-host systems are still lacking. We reviewed three recently developed generic pathway-oriented methods for modelling trade-related plant pest risks and their management. Two of these models (Holt et al., 2018; Jamieson et al., 2022) adopted a Bayesian network methodology. A third model (Jeger et al., 2018) was implemented as Monte Carlo simulations in the *Microsoft Excel @Risk™* add-on. All models were limited in their ability to support the evaluation of diverse risk-reducing measures, including their cumulative effects. Holt et al.'s (2018) model assessed the effect of selected measures on three qualitatively defined levels of pest infestation at successive control points in the production chain. The *Integrated Biosecurity Risk Assessment Model (IBRAM)* developed by Jamieson et al. (2022) ultimately assessed the risk of pest establishment for a given import pathway. IBRAM contained sub-modules for modelling risks of pest entry, escape, initial dispersal, establishment, spread and consequences, but there was limited scope for evaluating the effect of risk management. The Jeger et al. (2018) method also allowed for evaluating risks of pest entry, establishment, spread and impact associated with import pathways into the European Union. A range of 'risk-reducing options' could be incorporated via generalised model parameters that effect changes in the estimated pest quantities throughout the risk pathway.

Here, we describe and test the Pest Risk Reduction Scenario Tool (PRReSTo), a generic Bayesian network model that estimates likely infestation rates of host items in consignments. PRReSTo was developed to operationalise the functional risk framework proposed by van Klinken et al. (2020). It can in principle be applied to assessing and managing any trade-related plant pest risk resulting from infestation of vulnerable hosts during production. The model can inform pest risk assessment by explicitly considering host items' vulnerability to pest infestation and the pest abundance to which host items are exposed during production when calculating likely unrestricted host infestation rates. Foremost, PRReSTo supports pest risk management by enabling comparative evaluations of a comprehensive 'menu of risk-reducing measures' (van Klinken et al., 2022), individually or combined in a systems approach. We focus on a current implementation of the model, which was parameterised for quantifying infestation rates of horticultural produce by insect pests that can be trapped. This implementation is supported by an

open access web application. We apply the PRReSTo web app to common phytosanitary questions to test its versatility: evaluating likely (unrestricted) host infestation rates under different levels of pest exposure and host vulnerability; evaluating the benefit of measures that require pest monitoring with a corrective action; and comparing the individual and cumulative effects of common risk-reducing measures.

2. Generic model overview

2.1. Bayesian network approach

PRReSTo uses a Bayesian network (BN) approach, which has become a well-established area of research (Marcot and Penman, 2019) with many recent applications in biosecurity risk assessment, modelling and management (Smith et al., 2012; Holt et al., 2018; Froese et al., 2019; Jamieson et al., 2022; Meurisse et al., 2022). BNs are probabilistic models in which influencing variables (parent nodes) are linked by causal relationships to response variables (child nodes). Causal effects and interactions between multiple influencing variables can be graphically represented and quantitatively analysed based on underlying conditional probabilities (Fenton and Neil, 2013). Typically, each variable in a BN takes, or is discretised into, mutually exclusive categories (states) (Nojavan A et al., 2017). Based on knowledge or beliefs about the state of each influencing variable, the conditional probability that the response variable takes a certain state can be calculated. If the outcome (i.e., the state of the response variable) is known, the likely states of all influencing variables can similarly be inferred via Bayesian belief updating. Uncertainty (i.e., about the state of an influencing variable, or about causal relationships to a response variable) can be explicitly accommodated in BNs as probability distributions, and such uncertainty is propagated throughout the model (Nojavan A et al., 2017). PRReSTo was developed as a discrete BN using the graphical interface of the proprietary *Norsys Netica v.5.12* software (Appendix A). Like other authors (Holt et al., 2018; Froese et al., 2019; Jamieson et al., 2022), we support its current implementation with an open-access web app to facilitate adoption by decision-makers (Section 3.3).

2.2. Model structure

The PRReSTo model quantifies infestation rates of vulnerable host items resulting from infestation by a pest during production of a hypothetical consignment (or several consignments with a consistent risk profile) at a particular time and place. It has three functional parts, relating to the vulnerability of host items to pest infestation, the pest abundance to which vulnerable host items are exposed during production, and the resulting host infestation rates in the consignment (van Klinken et al., 2020, Fig. 1). Broadly, each part of the model consists of:

- (i) an initial expectation, or belief, about its likely unrestricted state in a given pest-host system,
- (ii) a series of functionally aligned risk-reducing measures (consistent with van Klinken et al., 2022), which can be applied at different efficacy levels during or after production, and
- (iii) a resulting estimate of its likely managed state given the initial expectation and the effects of any applied measures.

2.2.1. Exposure to pest

The exposure to pests when the host is vulnerable to infestation depends, first and foremost, on the pest abundance at the time and place of production. We use 'pest abundance' rather than 'pest prevalence' (e.g., IPPC 2017a) to unambiguously distinguish the abundance of pest populations in the field from pest infestation rates in consignments (Section 2.2.3). In PRReSTo, an **initial expectation** about unrestricted area-wide pest abundance in the production system is specified. This expectation can then be updated with observational evidence about pest abundance

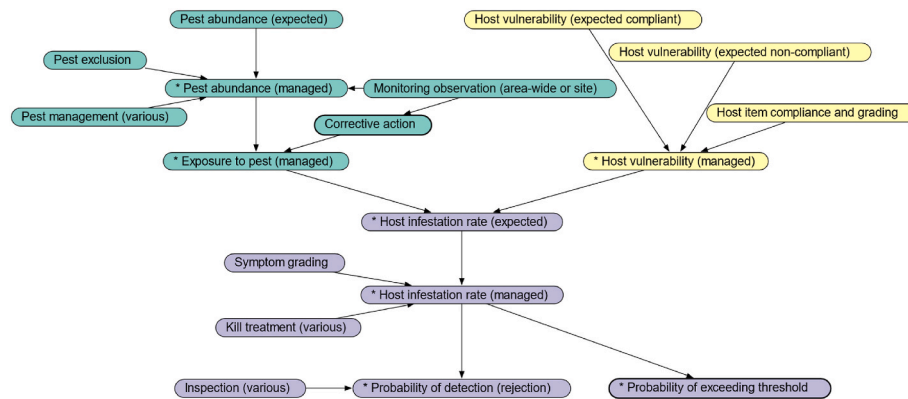


Fig. 1. A simplified representation of the PRReSTo model structure. The model has three functional parts consistent with the risk reduction objectives identified in van Klinken et al. (2020): exposure to pest (green), host vulnerability (yellow) and host infestation rate (purple). Response variables, each of which is connected to multiple influencing variables, are highlighted with an asterisk.

from monitoring or surveillance conducted at the area-wide scale or at a specific ‘registered site’. This evidence can increase confidence in **pest freedom** or **low pest abundance** claims if no or few pests are found. Pest abundance can also be mitigated via a range of measures. Measures that reduce pest abundance during production include **pest exclusion**, **pest management** or **corrective action** where observational evidence from monitoring exceeds an *a priori* action threshold. **Pest avoidance** measures such as limited spatial or seasonal overlap between host and pest can affect the initial expectation about area-wide pest abundance in the managed production system. Given all available information about pest abundance at the area-wide and site scales, and the effects of any risk-reducing measures, an **estimate of the likely managed pest abundance** to which host items in the consignment are exposed is obtained.

2.2.2. Host vulnerability

Host items in a consignment may vary in vulnerability depending on host quality and/or maturity stage. In PRReSTo, host vulnerability is defined by the relationship between pest abundance and host infestation rate (Section 3.1.2). **Initial expectations** about vulnerability are specified for host items that are compliant with **trade specifications** for quality and maturity stage (i.e., export grade) and those that are non-compliant (e.g., damaged, bruised or soft ripe). Where a percentage of host items in the consignment are assumed to be non-compliant, **maturity stage and quality grading** measures may be applied to remove them from the consignment. Given all available information about expected host vulnerability, host compliance with trade specifications, and the effects of any grading measures, an **estimate of the likely managed vulnerability** of host items in the consignment to infestation by the pest is obtained.

2.2.3. Host infestation rate

In PRReSTo, an **initial expectation** about infestation rates of host items in the consignment is calculated from the estimates of pest exposure and host vulnerability as described above (with or without risk-reducing measures). The expected infestation rates can then be updated with observational evidence about pest detection or non-detection from **inspection** conducted during production or post-production. Infestation rates in the consignment can also be reduced via a range of measures. These include **symptom grading** aimed at removing infested host items or various **kill treatments** aimed at killing, inactivating, or removing live pests from the consignment (van Klinken et al., 2022). Given all available information about levels of pest exposure and host vulnerability, and the effects of any risk-reducing measures, an **estimate of the likely managed host infestation rates** in the consignment is obtained.

3. Current implementation

The current implementation of PRReSTo was parameterised for quantifying infestation rates of vulnerable horticultural produce—measured as the percentage of infested host items (e.g., pieces of fruit) in a consignment—by insect pests that can be trapped. It has pragmatic, simplifying features and assumptions that are, however, appropriate for its main purpose as a versatile scenario tool. They may be empirically tested and improved upon through future research, as we discuss in Section 5. Each PRReSTo variable can take one, or in some cases a probability distribution across several, of a fixed set of discrete states (Table 1). For variables measuring pest abundance or host infestation rate, these discretised states cover the entire plausible parameter space, while each state was given a qualitative label (e.g., *very low*, *high*) (Section 3.1). For all other variables, a small but representative set of discrete states were defined. This limited model complexity while still allowing for broad exploration of the parameter space (e.g., for variables measured in percent, the interval [0,100]) via scenario analyses (Table 1). Assumptions for discretising variables and parameterising conditional probability relationships between influencing variables and response variables are briefly outlined below. The full model is provided in Appendix A and additional details in Appendix B.

3.1. Initial expectations (unrestricted state)

3.1.1. Pest abundance

In the current implementation of PRReSTo, we measured pest abundance using the proxy metric pests per trap per week (PTW). The metric was discretised into five intervals by orders of magnitude (i.e., log PTW) with a lower bound of zero and an upper bound of 100 PTW (Table 1; Appendix B). An initial expectation about unrestricted area-wide pest abundance in the production system is specified. Uncertainty (due to a lack of clear evidence) or variability (e.g., due to landscape heterogeneity, differences in production, or seasonal effects) may be modelled by specifying a probability distribution across several discretised PTW intervals. In the current implementation, this distribution may be either custom specified or calculated from the parameters μ (=mean) and σ (= standard deviation) of a normal distribution (both must be > 0), which is automatically truncated to the PTW range [0,100] (details in Appendix B).

3.1.2. Host vulnerability and infestation rate

In PRReSTo, we distinguished six levels of host vulnerability, each assigned a qualitative label ranging from *non-host* to *very high vulnerability* (Table 1). In the current implementation, an initial expectation about host vulnerability is specified for host items that are compliant with commercial or regulatory trade specifications for quality and/or

Table 1

The variables in the PRReSTo model, ordered by risk reduction objective. Each variable is defined and the discrete or discretised states it can take are listed. Definitions for variables relating to risk-reducing measures are summarised (with modifications) from [van Klinken et al. \(2022\)](#). *PTW = pests per trap per week.

Variable name	Variable definition	Discrete states
Exposure to pest		
Pest abundance (expected area-wide)	The expected unrestricted area-wide pest abundance (measured as PTW*) in the production system; may also be adjusted to implicitly model the effects of pest avoidance risk-reducing measures (e.g., limited spatial or seasonal overlap between host and pest) in a managed production system	- negligible (0–0.01 PTW) - very low (0.01–0.1 PTW) - low (0.1–1 PTW) - moderate (1–10 PTW) - high (10–100 PTW)
Pest exclusion	Risk-reducing measures aimed at partial or complete exclusion of the pest during production; may include physical barriers (e.g., protected facilities) and management practices (e.g., pest free inputs)	99% excluded 90% excluded 75% excluded 50% excluded - not applied
Pest management (various)	Risk-reducing measures aimed at minimising pest populations during production; may include agrochemicals (e.g., non-systemic cover sprays), attract and kill measures (e.g., pheromone traps, bait sprays), hygiene practices, biological control and other integrated pest management practices	99.9% effective 99% effective 90% effective 50% effective - not applied
Monitoring observation (area-wide or site)	The observational evidence about pest abundance (i.e., trap catch, measured as PTW) from monitoring or surveillance at either area-wide and/or site scales; may demonstrate efficacy of risk-reducing measures applied at either scale; may also be used to give confidence in pest freedom or low pest abundance if no or few pests are found	- negligible (0–0.01 PTW) - very low (0.01–0.1 PTW) - low (0.1–1 PTW) - moderate (1–10 PTW) - high (10–100 PTW) - no monitoring
Monitoring confidence	The confidence that the monitoring observation (i.e., trap catch) reliably reflects pest abundance across either the area-wide production system or registered site; may vary depending on the monitoring protocol (e.g., trap density, placement) and other factors (e.g., pest dispersal, habitat heterogeneity)	100% confident 75% confident 50% confident
Corrective action threshold	An <i>a priori</i> action threshold that, if exceeded by monitoring observations, requires application of a corrective action measure to reduce pest abundance below that threshold	>0.01 PTW (very low abundance) >0.1 PTW (low abundance) >1 PTW (moderate abundance) - not applied
Corrective action	Where an <i>a priori</i> action threshold is specified and exceeded by monitoring observations, the efficacy of the corrective action measure (e.g., bait spray) at reducing pest abundance below that threshold	99% effective 90% effective 75% effective 50% effective
Exposure to pest (managed consignment)	The pest abundance (measured as PTW) to which host items in the consignment are exposed given all available information about pest abundance at the area-wide and site scales, and the effects of any risk-reducing measures that have been applied	- negligible (0–0.01 PTW) - very low (0.01–0.1 PTW) - low (0.1–1 PTW) - moderate (1–10 PTW) - high (10–100 PTW)
Host vulnerability		
Host vulnerability (expected compliant)	The expected vulnerability to pest infestation of host items that are compliant with commercial or regulatory trade specifications for quality and maturity stage	- non-host - very low vulnerability - low vulnerability - moderate vulnerability - high vulnerability - very high vulnerability
Host vulnerability (expected non-compliant (mature stage or/low quality))	The expected vulnerability to pest infestation of host items that are non-compliant with commercial or regulatory trade specifications for quality (e.g., damaged, bruised) or maturity stage (e.g., soft ripe)	- no change +1 vulnerability level +2 vulnerability levels - very high vulnerability
Host item compliance (maturity stage/quality)	The percentage of host items in the consignment that are assumed to be compliant with trade specifications for quality or maturity stage	100% compliant 99% compliant 90% compliant 50% compliant - none compliant
Host item grading (maturity stage/quality)	Risk-reducing measure aimed at removing non-compliant (low quality or mature stage) host items from the consignment; may be applied during production or post-production	99% removed 90% removed 75% removed 50% removed - not applied
Host vulnerability (managed consignment)	The vulnerability of host items in the consignment to infestation by the pest given all available information about host vulnerability, host compliance with trade specifications, and the effects of any grading measures that have been applied	- non-host - very low vulnerability - low vulnerability - moderate vulnerability - high vulnerability - very high vulnerability
Host infestation rate		
Host infestation rate (expected consignment)	The expected unrestricted host infestation rates (measured as % of host items) in the consignment given the unrestricted, or managed where risk-reducing measures have been applied, estimates of pest exposure and host vulnerability	- negligible (<0.001%) - very low (0.001–0.01%) - low (0.01–0.1%) - moderate (0.1–1%) - high (1–10%) - very high (>10%)

(continued on next page)

Table 1 (continued)

Variable name	Variable definition	Discrete states
Symptom grading	Risk-reducing measure aimed at removing host items that are infested, or show symptoms of infestation, from the consignment; may be applied during production or post-production	• 99% removed • 90% removed • 75% removed • 50% removed • not applied
Kill treatment (various)	Risk-reducing measures aimed at killing, inactivating, or removing live pests from the consignment; may include agrochemicals, fumigation, heat/cold treatment, irradiation, surface cleaning, physical disturbance and other measures; may be applied during production (e.g., systemic pesticides), but frequently post-production (e.g., as an endpoint treatment)	• 99.9968% effective (probit 9) • 99.99% effective (probit 8.7) • 99.9% effective • 99% effective • 90% effective • 50% effective • not applied
Inspection sampling method	The application and sampling method of an inspection measure aimed at detecting pests in the consignment (and leading to rejection where a pest threshold is exceeded); may be applied during production (crop inspection) or post-production (e.g., packhouse or reject bin inspection) using a random sampling method, where relative risk = 1, or biased towards more vulnerable host items (e.g., damaged, bruised, soft ripe), where relative risk >1	• relative risk = 20 • relative risk = 10 • relative risk = 5 • relative risk = 2 • relative risk = 1 • not applied
Inspection sample size	Where an inspection measure is applied, the size of the sample (here n = number of host items inspected); may be pooled across repeat inspections of the registered site (e.g., crop inspections) or consignment (e.g., packhouse inspections)	• $n = 12,000$ • $n = 6,000$ • $n = 3,000$ • $n = 1,200$ • $n = 600$
Inspection sensitivity	Where an inspection measure is applied, the sensitivity of inspection to detecting a pest given it is present in an inspected host item; dependent on the type of host and pest, and inspection method and timing	• 100% sensitive • 90% sensitive • 50% sensitive • 20% sensitive
Probability of detection (rejection)	A binary representation of the observational evidence obtained from inspection; detection of the pest above an <i>a priori</i> threshold quantity (e.g., > zero pests detected) typically leads to rejection of the entire consignment	• ≥ 1 pests detected (+rejected) • no pest detected (+shipped)
Host infestation rate (managed consignment)	The host infestation rates (measured as % of host items) in the consignment given all available information about levels of pest exposure and host vulnerability, and the effects of any risk-reducing measures that have been applied to the consignment; can also be updated by observational evidence about pest detection or non-detection obtained through inspection	• negligible (<0.001%) • very low (0.001–0.01%) • low (0.01–0.1%) • moderate (0.1–1%) • high (1–10%) • very high (>10%)
Probability of exceeding threshold	A binary representation of the updated host infestation rates (measured as % of host items) in the consignment, summing the probabilities that it is below or above an <i>a priori</i> infestation rate threshold (e.g., 0.01%)	• below threshold (e.g., <0.01%) • above threshold (e.g., >0.01%)

maturity stage. An adjustment factor is selected for non-compliant low quality (e.g., damaged, bruised) or mature stage (e.g., soft ripe) host items (e.g., + 2 *vulnerability levels*; details in [Appendix B](#)).

Each vulnerability level was quantitatively defined by the relationship between pest abundance to which host items are exposed and host infestation rate. The greater the expected infestation rate at a given pest abundance, the more vulnerable the host. Infestation rate was measured as the percentage (%) of infested host items in the consignment and discretised into six intervals by orders of magnitude (i.e., log % infested; [Table 1](#)). In the current implementation, and in the absence of published empirical relationships, we described the six host vulnerability levels using different hypothetical log-log linear relationships between log PTW and log % infested, with a common intercept = 0 and varying slopes ([Fig. 2](#)). After plotting these lines, we identified their intersections with each pest abundance interval's upper and lower boundaries. For each combination of host vulnerability level and pest abundance interval, this allowed us to calculate conditional probabilities that the expected infestation rate falls within each of its discretised intervals by apportioning the corresponding line segment to interval bins. This implementation captures some uncertainty if line segments extend across more than one infestation rate interval (see example in [Fig. 2](#); details in [Appendix B](#)).

3.2. Risk-reducing measures

PRReSTo was developed so that most of the commonly used measures influencing host infestation rates listed in [van Klinken et al. \(2022\)](#) can be captured in a scenario analysis ([Fig. 3](#)). Most measures are modelled directly using the corresponding PRReSTo variable, or in some

cases several variables ([Table 1](#); [Fig. 3](#)). We describe their current implementations below. Some measures are currently modelled implicitly by adjusting the initial expectations about pest abundance (e.g., pest avoidance measures to limit the spatial or temporal overlap between host and pest) or host vulnerability (e.g., measures that limit trade to less vulnerable hosts or parts of hosts) to reflect their likely managed states ([Fig. 3](#)).

3.2.1. Site management and consignment treatment

Variables for pest management and pest exclusion measures were parameterised based on their efficacy in reducing pest abundance during production ([Table 1](#); details in [Appendix B](#)). Similarly, variables for measures that affect live pests in host items or remove infested host items from the consignment were parameterised based on their efficacy in reducing host infestation rate (i.e., % of host items infested) ([Table 1](#); [Appendix B](#)). The same PRReSTo variable may be used for different measures in the same measure category and irrespective of when the measure was applied, for example whether a kill treatment was applied during or post-production ([Fig. 3](#); [van Klinken et al., 2022](#)). We defined generic states that reflect typical efficacy levels achieved in practice while also allowing flexibility for exploring a range of plausible scenarios ([Table 1](#)). Where several functionally aligned measures are applied together (e.g., up to two pest management and a pest exclusion measure), their efficacy levels were multiplied when calculating post-measure pest abundance or host infestation rate.

3.2.2. Pest monitoring and corrective action

In PRReSTo, the initial expectation about area-wide pest abundance may be updated with observational evidence from monitoring or

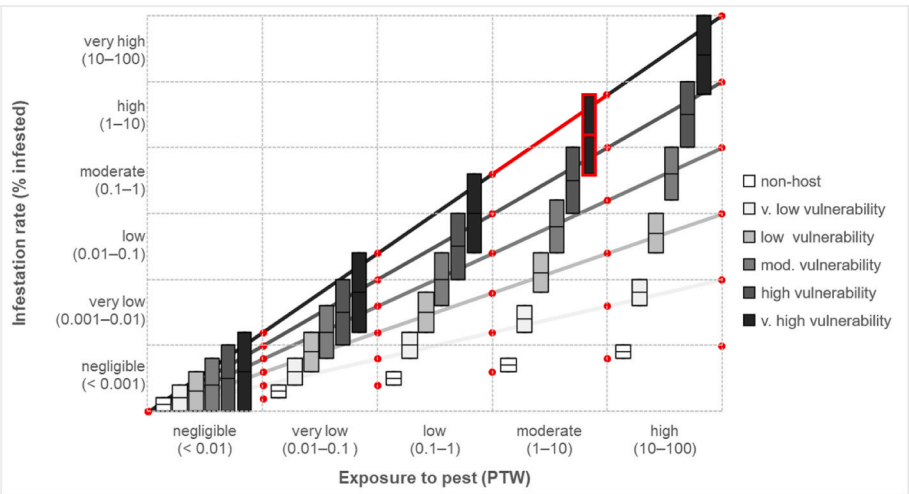


Fig. 2. Expected host infestation rates (% infested) given different combinations of host vulnerability and pest abundance to which the consignment is exposed. For each host vulnerability level, a different underlying log-log linear relationship between log PTW and log % infested was defined and discretised into line segments, one for each pest abundance interval (x axis). The proportion of a line segment within each infestation rate interval (y axis) was then calculated and used as conditional probabilities for the expected infestation rate. Here we display line segments as colour-coded bars for ease of visualisation. For example (red line segment and bar), given host vulnerability = *very high* and pest abundance = *moderate* (1–10 PTW), the line segment/bar extends across the two infestation rate intervals *high* (1–10%) and *moderate* (0.1–1%), with the resulting conditional probabilities calculated as 67% and 33% respectively.

RISK REDUCTION OBJECTIVE	CONSIGNMENT STAGE				
	Production		Post-production	Post-border	
	Minimise exposure to pests when the host* is vulnerable			Reduce establishment risks	
	PEST FREEDOM OR LOW PEST ABUNDANCE** - Use Monitoring observation (area-wide) and Monitoring efficacy (confidence) to model Area-wide - Use Monitoring observation (site) and Monitoring efficacy (confidence), in conjunction with Corrective action threshold and Corrective action, to model Registered site	PEST MANAGEMENT Use Pest management (various) to model any of (up to two at a time): Agrochemicals Attract and kill Biological control Hygiene Sanitation Other pest management tools Integrated pest and disease management	PEST AVOIDANCE - Adjust initial expectation for Pest abundance (expected area-wide) to implicitly model Restricted to poor pest habitat Limit seasonal overlap Limit exposure time to pest Isolation from hot spots - Habitat manipulation is rarely used and currently not implemented	PEST EXCLUSION - Use Pest exclusion to model Protected facilities Safeguarding Protected units Pest-free inputs - Adjust initial expectation for Pest abundance (expected area-wide) to implicitly model Maintain buffer zone - Segregation is rarely used during production and currently not implemented	LIMIT PROPAGULE PRESSURE - Trade volume Consignment and packaging size Prevent escapes are currently not implemented LIMIT EXPORT DESTINATIONS OR USE - Restricted to poor pest habitat Poor time of year Restricted end-use are currently not implemented
	Minimise vulnerability of the host* to infestation				
POOR HOST OR CARRIER Adjust initial expectations for Host item vulnerability (expected trade specifications) to model Poor host or carrier status Modify vulnerability Remove/prohibit parts of host	- Use Host item compliance (maturity stage), in conjunction with Host item grading (maturity stage) to model Poor developmental stage - Use Host item compliance (quality) in conjunction with Host item grading (quality) to model Quality specifications				
Reduce infestation rates					
REDUCE PEST IN COMMODITY - Use Kill treatment (various) to model any of (up to two at a time): Treatments (to kill or inactivate the pest) Physical disturbance and processing Surface cleaning - Remove contaminants is currently not implemented	REMOVE INFESTED COMMODITY UNITS - Use Symptom grading to model Symptom grading - Risk profiling is rarely used and currently not implemented.		REMOVE INFESTED CONSIGNMENT Use Inspection sampling method, Inspection sample size and Inspection sensitivity to model Inspect product and reject Quarantine and reject		

Fig. 3. Explanation of how risk-reducing measures listed in van Klinken et al. (2022) can be modelled using the PRRestO variables described in Table 1 (red). Measures (italics) are grouped into measure categories (capitalised) under each of the four risk reduction objectives, which in turn are aligned under the most relevant consignment stages. We use the terms ‘host’ rather than ‘commodity’ (*) and ‘pest abundance’ rather than ‘pest prevalence’ (**) as used in van Klinken et al. (2022).

surveillance conducted during production at area-wide or site scales. Monitoring is required to support pest free area/site (IPPC 2016a; IPPC 2017b) or area of low pest prevalence (IPPC 2016b) measures (Fig. 3). It may also demonstrate the effectiveness of site management measures. In the current implementation, trap catches observed at either scales are entered by selecting one of the five discretised log PTW intervals (Table 1). However, an observed trap catch may not reliably reflect pest

abundance across the area-wide production system or at the registered site. Confidence in reliability may be affected by the monitoring protocol (e.g., trap density, placement) and other factors (e.g. pest dispersal, habitat heterogeneity) (van Klinken et al., 2023). We distinguished three generic levels of confidence in the specified monitoring observation (Table 1). These affect how the initially expected area-wide pest abundance is updated in the model. For example, if pest abundance was

expected to be *high* and a *low* site trap catch was observed with 50% confidence, a 50% probability that pest abundance at the site is in fact higher than the observed trap catch remains (Appendix B).

Irrespective of confidence in monitoring observations, if an *a priori* action threshold (e.g., 0.1 PTW) is exceeded, a corrective action measure (e.g., bait spray) may be triggered to reduce pest abundance below that threshold (Fig. 3). In PRReSTo, either of three alternative thresholds may be specified (Table 1). If a corrective action threshold is specified and exceeded by monitoring observations, the efficacy of the corrective action at reducing pest abundance below that threshold must also be specified (Table 1; Fig. 3; Appendix B).

3.2.3. Inspection (detect and reject)

While monitoring aims to estimate pest abundance at the time and place of production, inspection measures aim to detect infested host items. Inspection may be conducted during production (e.g., crop inspection) or post-production (e.g., packhouse inspection). Typically, it leads to suspension of a registered site or rejection of a consignment where an *a priori* detection threshold is exceeded (Fig. 3). In PRReSTo, we measured the probability of a positive inspection outcome (i.e., ≥ 1 pest detected) using the internationally agreed methodology described in ISPM31 (IPPC 2016c). This considers sample size, inspection sensitivity and host infestation rate. We also included a relative risk parameter which may be used to model the effect of different sampling methods (details in Appendix B). A representative set of discrete states for each inspection parameter are currently implemented in PRReSTo (Table 1).

3.2.4. Host item specifications and grading

In PRReSTo, following specification of likely vulnerability levels for compliant and non-compliant host items (Section 3.1.2), the effect of measures aimed at limiting trade to less vulnerable hosts or parts of hosts is modelled using two variables (Fig. 3). First, the percentage of host items in the consignment that are assumed to be compliant with trade specifications for quality and/or maturity stage is specified. This may be used to indicate the efficacy of trade specification measures (e.g., 90% compliant). Next, the efficacy of host item grading measures at removing non-compliant (low quality or mature stage) host items from the consignment, may also be specified (e.g., 90% removed; Table 1; details in Appendix B).

3.3. Web application and model outputs

We translated the current implementation of PRReSTo into R code (version 4.2.1; R Core Team 2022) and made it freely available as a Shiny (Chang et al., 2022) web app (Beeton et al., 2022; <https://shiny.csiro.au/PRReSTo/>) without needing to interact with the underlying BN model (provided as Appendix A). The web app user must first specify initial expectations about unrestricted area-wide pest abundance and host vulnerability (Section 3.1). All model variables associated with risk-reducing measures are defaulted to their *not applied* state (Section 3.2), but any of the discrete states as they are currently defined (Table 1) may be selected.

Three model outputs are presented:

- The host infestation rate in the consignment as a probability distribution across all six infestation rate intervals; this takes into account all user-supplied information about levels of pest exposure and host vulnerability, and the effects of any risk-reducing measures that have been applied to the consignment. If an inspection measure has been applied, the user may also choose to display likely host infestation rates given a negative inspection outcome (i.e., *no pest detected and consignment shipped*).
- The probability that a user-defined *a priori* acceptability threshold (e.g., 0.01% host infestation rate) is exceeded in the consignment (see also Table 1).

- The probability of a positive inspection outcome (i.e., ≥ 1 pest detected and consignment rejected); if no inspection measure is applied, this is zero.

4. Case study applications

We apply the PRReSTo web app to three hypothetical case studies to test its versatility as a scenario analysis and decision support tool for common pest risk assessment and pest risk management questions faced by industries and regulators.

4.1. Evaluating unrestricted host infestation rates (risk assessment function)

4.1.1. Use case

The primary aim of pest risk assessment is to assess unrestricted risk in a pest-host system for the purposes of trade. While this requires consideration of trade pathways through to the likelihood and consequences of pest establishment and spread (IPPC 2017c), likely infestation rates of host items in consignments are a key component, and the main focus of pest risk management. Here, we used PRReSTo to evaluate likely (unrestricted) host infestation rates in consignments under alternative initial expectations about (unrestricted) area-wide pest abundance and host vulnerability.

4.1.2. Methods

We described the expected pest abundance using a truncated normal distribution with parameters μ and σ for two hypothetical scenarios: *few pests* ($\mu = 0.001$, $\sigma = 0.01$) and *many pests* ($\mu = 1$, $\sigma = 1$). PRReSTo automatically calculates the probability for each discretised pest abundance interval. We also evaluated three different host vulnerability levels (*very low*, *moderate* or *very high*).

4.1.3. Results

If the host items in consignments are exposed to *few pests* during production, host infestation rates were most likely *negligible* ($< 0.001\%$ or $< 1:100,000$ infested) in all vulnerability scenarios. Yet, if host vulnerability is *very low* that outcome was certain compared to a degree of uncertainty if vulnerability is *moderate* or *very high*. With an *a priori* acceptability threshold of 0.01% (i.e., 1:10,000) infestation rate, there was an 11% chance that this threshold is exceeded in the *very high* vulnerability scenario (Fig. 4a).

If host items are exposed to *many pests* and their vulnerability is *very high*, then there was an 40% chance of *high* (1–10%) and 38% chance of *moderate* (0.1–1%) infestation rates in the consignment. At lower vulnerability levels, likely infestation rates decreased by at least one order of magnitude to most likely (63% chance) *low* (0.01–0.1%) under *moderate* vulnerability, and to certainly below the 0.01% infested acceptability threshold under *very low* vulnerability (Fig. 4b).

4.1.4. Conclusions

Clearly, likely (unrestricted) host infestation rates vary significantly depending on both host vulnerability and pest exposure during production. In some scenarios they were likely *negligible* or *very low* even without any risk-reducing measures. Looking at the relationships in terms of an *a priori* acceptability threshold is useful, but also conceals important nuance that can only be appreciated via a full probability distribution across host infestation rates. The PRReSTo web app displays results in both ways.

4.2. Evaluating pest monitoring and corrective action

4.2.1. Use case

Confidence in low pest abundance underpins many pest risk management protocols (van Klinken et al., 2020). One of the most common risk-reducing measures requires regular monitoring or surveillance at

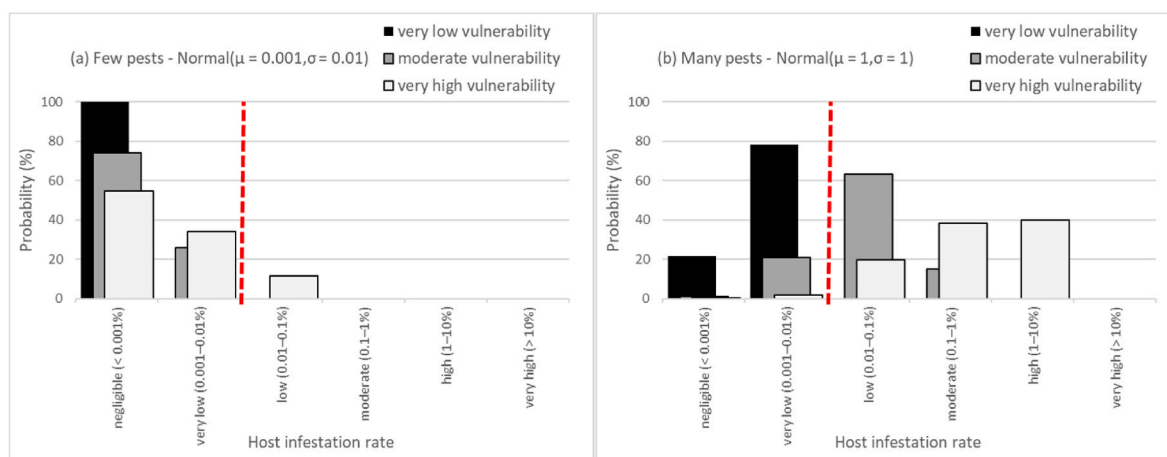


Fig. 4. Likely infestation rates for host items of *very low* (black), *moderate* (dark grey) or *very high* (light grey) vulnerability under two alternative pest abundance scenarios (a = *few pests* and b = *many pests* as described by a normal distribution with parameters μ and σ). We show the probability distribution across all six discretised host infestation rate intervals. The dotted red line highlights a 0.01% (i.e., 1:10,000) infestation rate threshold.

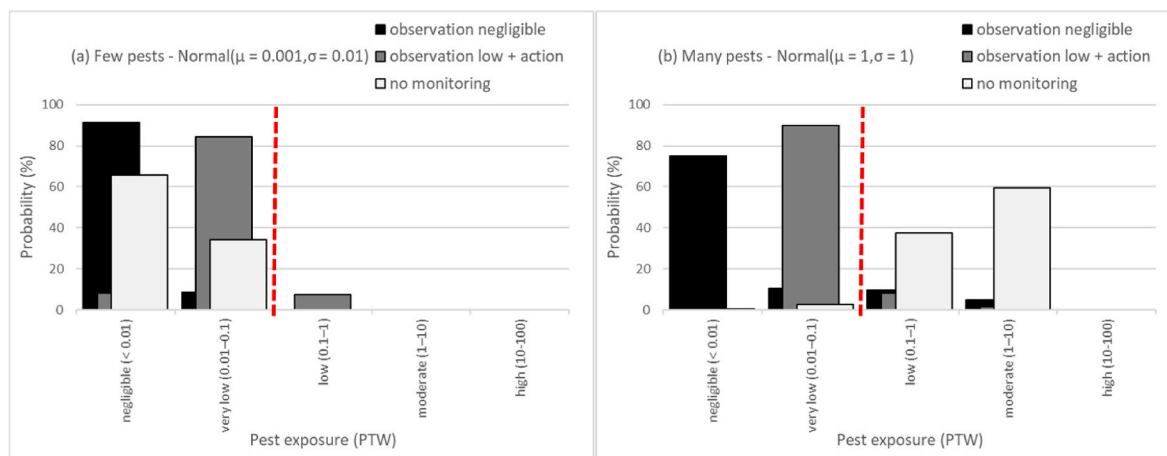


Fig. 5. Likely pest exposure of consignments for different monitoring outcomes—no monitoring (light grey); monitoring results in a *low* (0.1–1 PTW) trap catch and corrective action is taken (dark grey); monitoring results in a *negligible* (< 0.01 PTW) trap catch (black)—under two alternative expected pest abundance scenarios (a = *few pests*; b = *many pests* as described by a normal distribution with parameters μ and σ). We show the probability distribution across all five discretised pest abundance intervals. The dotted red line highlights a 0.1 PTW action threshold.

area-wide and/or site scales to support claims of “pest free area/site” (IPPC 2016a; IPPC 2017b) or “area of low pest prevalence” (IPPC 2016b) status. If such monitoring results in a trap catch above an *a priori* threshold, there is a consequence (typically either a corrective action or trade suspension of the registered site). Here, we used PRReSTo to evaluate whether and to what extent pest monitoring and corrective action measures may reduce the likely pest exposure of consignments.

4.2.2. Methods

We investigated three scenarios under the same two initial pest abundance expectations defined in Section 4.1: *few pests* ($\mu = 0.001$, $\sigma = 0.01$) and *many pests* ($\mu = 1$, $\sigma = 1$). A baseline scenario with no monitoring was compared to two risk reduction scenarios (both assumed 75% confidence in the monitoring protocol and a corrective action threshold > 0.1 PTW but differed in the monitoring observation obtained):

- observation negligible: monitoring results in a *negligible* (0–0.01 PTW) trap catch
- observation low + action: monitoring results in a *low* (0.1–1 PTW) trap catch, therefore exceeding the > 0.1 PTW threshold and triggering a corrective action measure (e.g., bait spray; here assumed to be 90% effective at reducing pest abundance below that threshold)

4.2.3. Results

If *few pests* are expected to be present during production, confirming this expectation by a *negligible* (0–0.01 PTW) trap catch increased the probability that pest exposure is in fact *negligible* from 66% to 92% (Fig. 5a). This effect also held if *many pests* are expected (probability increased from 0% to 75%), although there remained a greater probability that pest exposure is higher than the monitoring observation indicates (15% chance that it is above the action threshold 0.1 PTW; Fig. 5b). Of course, the probability of above-threshold pest abundance would be lower if a higher confidence in monitoring observations was assumed.

Conversely, if monitoring results in a *low* (0.1–1 PTW) trap catch and a corrective action is consequently triggered, the initial expectation about area-wide pest abundance had little effect on results: pest abundance was above the 0.1 PTW action threshold with 8% probability in either scenario (Fig. 5). This would decrease if corrective action efficacy was higher than the modelled 90% (but this may be difficult to achieve, as time delays before the corrective action takes effect need to be considered as well as kill rates).

4.2.4. Conclusions

Measures involving pest monitoring and corrective action can reduce risk in two ways. Catching no or very few pests can provide confirmatory evidence for expected below-threshold pest abundance. Above-threshold trap catches trigger a corrective action, which in turn reduces the pest abundance to which host items in consignments are exposed to below-threshold levels. A well-informed initial expectation about area-wide pest abundance (e.g., from empirical data collected in production and non-production areas) is valuable when confidence in site-based monitoring observations is low. Production in areas with significant pest pressure requires well-designed monitoring protocols to give confidence that pest exposure is below a desired threshold.

4.3. Comparatively evaluating measures (risk management function)

4.3.1. Use case

Where unacceptable risks are identified through pest risk assessment, trading partners may require those risks to be managed through the application of phytosanitary measures. In many cases, this is an individual measure with a demonstrated mortality rate (e.g., a probit 8.7 or probit 9 single-point treatment). Alternatively, multiple measures are combined into phytosanitary systems approaches (van Klinken et al.,

2020). Here, we used PRReSTo to comparatively evaluate the individual and cumulative effects of common risk-reducing measures under different host infestation scenarios.

4.3.2. Methods

Likely (unrestricted) host infestation rates (with no risk-reducing measures applied) were estimated for the *many pests* ($\mu = 1$, $\sigma = 1$) pest abundance scenario, and the same three alternative host vulnerability levels (*very low*, *moderate* or *very high*) as in Section 4.1. We evaluated nine commonly cited measures, or combinations of measures (van Klinken et al., 2022; see details in Appendix C). For simplicity, we report risk-reducing effects as the probability that either of two *a priori* acceptability thresholds—0.01% (1:10,000 infested) or 0.001% (1:100,000 infested)—were exceeded rather than as full probability distributions across host infestation rates.

4.3.3. Results

If the host items in a consignment are exposed to *many pests* during production with no risk-reducing measures, host infestation rates very likely exceeded the 0.001% threshold irrespective of host vulnerability. Hosts with *very high* and *moderate* vulnerability also likely exceeded the 0.01% threshold (98% and 78% chance respectively, Fig. 6). Non-detection of a pest during an inspection of 600 packed host items with a sensitivity of 100% hardly affected these results. This is because infested host items are rarely detected with such inspection parameters when infestation rates are below 0.1%.

Field monitoring that results in a *negligible* (< 0.01 PTW) trap catch reduced the probability of infestation rates exceeding either acceptability threshold (Fig. 6). Residual above-threshold probabilities were due to the assumed limited 75% confidence in the observed *negligible* trap catch, given the initial expectation that there are *many pests* (see Fig. 5). Similar risk reduction was achieved by a limited seasonal overlap measure, assuming that it reduces the probability that pest abundance is *low* (> 0.1 –1 PTW) to just 1% (and that it is *very low* (0.01–0.1 PTW) or *negligible* (0–0.01 PTW) to 49.5 % each; Appendix C). If the two measures are combined—i.e., an expected limited overlap between host and pest is additionally confirmed by an observed *negligible* (< 0.01 PTW) trap catch, then—host infestation rates were unlikely (0–10% chance) to exceed either acceptability threshold. The one exception was *very highly* vulnerable hosts when the acceptability threshold was $< 0.001\%$ (Fig. 6).

Applying pesticides during production can reduce host infestation rates by managing pest populations in the field or by killing live pests in or on infested host items (van Klinken et al., 2020). A 90% effective systemic pesticide that acts in both ways had a much greater risk-reducing effect than a non-systemic cover spray that kills 90% of the pest population but does not affect live pests in infested host items. For example, for hosts with *very low* vulnerability a systemic pesticide reduced the probability of infestation rates exceeding either acceptability threshold to zero. For *moderately* vulnerable hosts, the probability of infestation rates exceeding the 0.01% threshold was also reduced to zero, but not the probability of exceeding the 0.001% threshold (Fig. 6).

A post-production single-point treatment that kills 99% of live pests in host items had a similar risk-reducing effect to the less effective (90%) but dual action systemic pesticide applied during production. However, application of a highly effective (99.99%, equivalent to a probit 8.7 level of protection) kill treatment reduced the probability that host infestation rates exceed either acceptability threshold to zero in all scenarios. Combining a 99% effective post-production kill treatment with a limited seasonal overlap measure during production was another way for ensuring below-threshold host infestation rates irrespective of host item vulnerability (Fig. 6).

4.3.4. Conclusions

Our analysis shows that the adequacy of risk-reducing measures is highly dependent on the level of unrestricted host infestation in a pest-

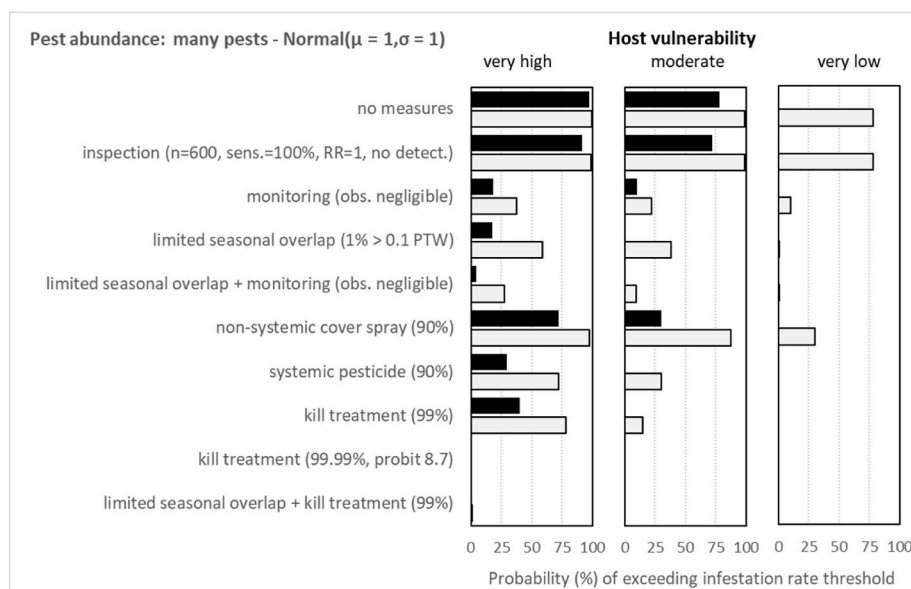


Fig. 6. The effect of commonly used risk-reducing measures on the probability of host infestation rates exceeding either of two *a priori* acceptability thresholds—0.01% (1:10,000 infested; black) and 0.001% (1:100,000 infested; light grey). We show results for host items of very low, moderate or very high vulnerability under a *many pests* pest abundance scenario.

host system, and on what residual host infestation rate is considered acceptable (here defined via two *a priori* thresholds). Where unrestricted infestation rates are likely high (*many pests* and *very highly* vulnerable hosts) and the managed host infestation rates must be *negligible* (below the < 0.001% acceptability threshold), adequate risk reduction may be achieved by highly effective individual measures (e.g., a probit 8.7 equivalent kill treatment) or combinations of several less effective measures (e.g., a 99% effective kill treatment + limited seasonal overlap) in a systems approach. Adequate risk reduction is more easily achieved if hosts have moderate or very low vulnerability.

5. Discussion

Despite recent advances in the development of quantitative approaches to assessing and managing trade-related plant pest risks, agreed methods and user-friendly tools for evaluating diverse risk-reducing measures in the context of different pest-host systems are lacking. Here, we described the Pest Risk Reduction Scenario Tool (PRReSTo), a generic Bayesian network model that estimates likely infestation rates of host items in consignments. The model builds on the functional risk framework recently developed by van Klinken et al. (2020). PRReSTo can inform both pest risk assessment and pest risk management of trade-related plant pest risks resulting from infestation of vulnerable hosts during production. Specifically, the tool allows for comparatively evaluating the risk-reducing effects of a comprehensive 'menu of measures' during production and post-production (van Klinken et al., 2022), individually or combined in a systems approach. The current implementation of PRReSTo, while limited to quantifying infestation rates of horticultural produce by insect pests that can be trapped, is supported by an open access web application to facilitate adoption by industry and regulators. We used the PRReSTo web app to investigate common phytosanitary questions. Analyses demonstrated its versatility as a tool for quickly and easily assessing likely (unrestricted) host infestation rates under different levels of pest exposure and host vulnerability, and for assessing the adequacy of risk-reducing measures under different host infestation scenarios and acceptability thresholds. Where production occurs in areas with high pest abundance, scenario analysis suggests that equivalent risk reduction may be achieved by a probit 8.7 post-production treatment or a combination of measures in a systems approach.

Under the SPS Agreement, WTO members are required to apply risk-based decision-making to the assessment of plant health risks posed by international trade, and the selection of measures needed to manage identified risks to an Appropriate Level of Protection (ALOP) (WTO 1994; IPPC 2006). A lack of agreed methods for assessing risk against alternative protocols (Jamieson et al., 2014; van Klinken et al., 2020) has contributed to an over-reliance on single-point treatments with set mortality criteria (typically probit 8.7 or 9.0) (Follett and Neven 2006). The widespread use of such treatments irrespective of the level of unrestricted risk determined through pest risk assessment can result in "widely varying risks of an introduction occurring and does not give meaningful levels of security" (Landolt et al., 1984). It can result in protocols that poorly balance the need for effective risk management with the need for least trade restrictive impositions (Devorshak 2012). The functional risk framework (van Klinken et al., 2020) and menu of measures (van Klinken et al., 2022) categorised measures according to how they reduce risk. This provided the basis for developing and assessing risk-based protocols, including phytosanitary systems approaches. By operationalising the risk framework in a generic model, PRReSTo allows users to quantitatively assess and compare likely host infestation rates under different unrestricted (including, potentially, commercially managed) and regulated scenarios. The tool may support trading partners in more transparently developing and assessing the equivalence of alternative protocols for meeting ALOP, as required under the SPS agreement.

There are other generic quantitative methods for assessing and managing plant pest risks, most notably the pathway-oriented models published in Jeger et al. (2018), Holt et al. (2018) and Jamieson et al. (2022). PRReSTo is the first tool that allows users to evaluate the individual or cumulative effects of a comprehensive "menu of measures" in a functional risk reduction framework (van Klinken et al., 2022) through an open access web-interface. Yet, there are several areas where PRReSTo could be improved to better align or integrate with pathway approaches. First, PRReSTo currently estimates a relative probability that host items in hypothetical consignments are infested to a certain percentage and provides uncertainty around this estimate. Some pathway models (Hattingh et al., 2020; Jamieson et al., 2022) provide more informative absolute estimates of pest infestation (e.g., number of pests in consignments and/or number of consignments infested). To address this limitation, information on infestation size (number of pests

per host item), consignment size (number of host items per consignment) and overall trade volume (number of consignments shipped) would need to be integrated into the PRReSTo model. Second, PRReSTo is currently limited to quantifying likely host infestation rates in shipped consignments. Assessing and managing risk in accordance with international PRA guidelines (Devorshak 2012; IPPC 2007; IPPC 2017c) requires linking host infestation in consignments through to the likelihood and consequences of pest establishment and spread following import. Extending PRReSTo to estimate the likelihood of pest establishment is a priority for future development. This would require estimation of absolute pest numbers and the addition of variables associated with quantifying and reducing establishment (Fig. 3; van Klinken et al., 2022). Alternatively, PRReSTo's estimates of likely host infestation rates in consignments, with and without commercial or regulated management, could be used as inputs into more complete pathway models. Third, the IBRAM (Jamieson et al., 2022) and EFSA (Jeger et al., 2018) pathway models treat all model parameters as continuous variables. In contrast, PRReSTo is currently implemented as a discrete BN, which meant that the inherently continuous response variables pest abundance or host infestation rate needed to be discretised into broad intervals by orders of magnitude. For all other variables, a small but representative set of discrete states were defined. Discretisation inevitably leads to information loss, model error and ambiguity in the interpretation of results (Nojavan A et al., 2017). While continuous BNs are a developing research area (Marcot and Penman 2019), this is a priority for future implementations of the PRReSTo model.

Uncertainty about pest exposure, host vulnerability, infestation rates and the efficacy of diverse risk-reducing measures is an inherent problem for pest risk assessment and management. In their recent analysis of published pest risk management protocols, van Klinken et al. (2022) found large variability in the empirical support for risk-reducing measures. We designed PRReSTo primarily as a scenario analysis tool that allows users to explore different beliefs or expectations about the pest-host system and comparatively evaluate alternative risk management options. Uncertainty and spatial or temporal variability in the production system may also be captured and propagated through the model. Where data or knowledge is available to directly inform initial expectations about area-wide pest abundance and host vulnerability or beliefs about measure efficacy, this may be used to specify model inputs. We deliberately used simple rules for parameterising the conditional probability relationships between influencing variables and response variables underlying the PRReSTo model. For example, for each of the defined generic levels of measure efficacy (e.g., 50%, ..., 99.9%, Table 1), we used the corresponding pest survival rates (e.g., 0.5, ..., 0.001) to calculate likely post-measure pest abundance or host infestation rates under simplifying assumptions (Appendix B).

A key relationship in the PRReSTo model is the interaction between host vulnerability to pest infestation and pest exposure of a vulnerable hosts to determine likely host infestation rates. This is an important advance from previous attempts at operationalising the host vulnerability concept for PRAs (e.g., Follett et al., 2021). However, the currently defined log-log linear relationships between PTW (pests per trap per week), a proxy metric of pest abundance, and resulting infestation rates for each host vulnerability level are hypothetical. They are also quite limited in how uncertainty is captured (Fig. 2). They were formulated as useful initial hypotheses whose effect may then be explored via scenario analysis. Furthermore, although metrics of trap catch such as PTW are widely recommended in pest risk management (FAO/IAEA et al., 2018; IPPC 2016b), empirical studies on the relationship between trap catch and host infestation are limited. For example, Grechi et al. (2021) found a weak relationship, but for a much narrower parameter range than that implemented in PRReSTo: the ~3–12% infestation rates observed in Grechi et al. (2021) would mostly be in the *high* (1–10%) infestation rate interval. Allowing for greater flexibility in modifying the relationship between pest abundance and host infestation rate is a priority for future implementations of the PRReSTo model. This should allow for specification of

uncertain non-linear relationships, and how these may be affected by trap efficacy. Ultimately, the validity of any such relationships should be empirically tested and, where sufficient data is available, fitted for specific pest-host systems.

The generic PRReSTo model, which operationalises the risk framework proposed by van Klinken et al. (2020), may in principle be applied to assessing and managing any trade-related plant pest risk. The current implementation of PRReSTo, which is supported by an open access web app and has been detailed and tested in this paper, was developed specifically for insect pests of horticultural produce that can be trapped. However, the generic model could be readily adapted to alternative applications with some changes to the current model parameterisation. For example, the unit for measuring pest abundance (PTW) would need to be altered for some pests and pathogens. Similarly, the unit for measuring infestation rate (% of host items infested) may need to be altered for commodities such as grain. We anticipate that the application of PRReSTo will provide regulators, industry and researchers with greater confidence when assessing and managing trade-related plant pest risks, and stimulate the development and trialling of innovative, risk-based approaches to decision-making.

Declaration of competing interest

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Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cropro.2023.106484>.

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